Why replay works: linking the diagnosis to the benchmark result

Diagnosis (§6.1)

Forgetting concentrates at the classifier head and late layers. The head dominates Fisher importance (2.2×10^{-1}) and drifts most (CKA 0.215); the input encoders stay stable.

Mechanism: experience replay

A small buffer of past-task examples is replayed at every step. Old-class targets keep producing gradient at the output head, so the head is not overwritten as new tasks arrive.

Outcome (§6.2)

The head stays aligned with earlier tasks. On DIL: AA 88.2 and BWT ≈ 0 — matching the Joint oracle (84.8); forgetting is essentially eliminated.

Hypothesis (open): because the forgetting locus is output-facing, a mechanism that targets the head / late layers directly — the proposed DocCL — could match replay without storing past data. To be tested once the method is finalised.

Why not the others? EWC constrains all weights broadly → partial help, mainly under domain shift. LwF distils old-class logits that decay as the head grows → no measurable benefit in dense token classification.